A Technical Introduction to Restricted Boltzmann machine

Section 1

P (h j = 1 | v) = σ (b j + ∑ i = 1 m w i , j v i) {\displaystyle P(h_{j}=1|v)=\sigma \left(b_{j}+\sum _{i=1}^{m}w_{i,j}v_{i}\right)} and P (v i = 1 | h) = σ (a i + ∑ j = 1 n w i , j h j) {\displaystyle \,P(v_{i}=1|h)=\sigma \left(a_{i}+\sum _{j=1}^{n}w_{i,j}h_{j}\right)}

Section 2

where σ {\displaystyle \sigma } denotes the logistic sigmoid. The visible units of Restricted Boltzmann Machine can be multinomial, although the hidden units are Bernoulli. In this case, the logistic function for visible units is replaced by the softmax function

Section 3

Structure The standard type of RBM has binary-valued (Boolean) hidden and visible units, and consists of a matrix of weights W {\displaystyle W} of size m × n {\displaystyle m\times n} . Each weight element (w i , j) {\displaystyle (w_{i,j})} of the matrix is associated with the connection between the visible (input) unit v i {\displaystyle v_{i}} and the hidden unit h j {\displaystyle h_{j}} . In addition, there are bias weights (offsets) a i {\displaystyle a_{i}} for v i {\displaystyle v_{i}} and b j {\displaystyle b_{j}} for h j {\displaystyle h_{j}} . Given the weights and biases, the energy of a configuration (pair of Boolean vectors) (v,h) is defined as

Section 4

E (v , h) = − ∑ i a i v i − ∑ j b j h j − ∑ i ∑ j v i w i , j h j {\displaystyle E(v,h)=-\sum _{i}a_{i}v_{i}-\sum _{j}b_{j}h_{j}-\sum _{i}\sum _{j}v_{i}w_{i,j}h_{j}}

Section 5

A restricted Boltzmann machine (RBM) (also called a restricted Sherrington–Kirkpatrick model with external field or restricted stochastic Ising–Lenz–Little model) is a generative stochastic artificial neural network that can learn a probability distribution over its set of inputs. RBMs were initially proposed under the name Harmonium by Paul Smolensky in 1986, and rose to prominence after Geoffrey Hinton and collaborators used fast learning algorithms for them in the mid-2000s. RBMs have found applications in dimensionality reduction, classification, collaborative filtering, feature learning, topic modelling, immunology, and even many■body quantum mechanics.

Section 6

Restricted Boltzmann machines are trained to maximize the product of probabilities assigned to some training set V {\displaystyle V} (a matrix, each row of which is treated as a visible vector v {\displaystyle v}),

Section 7

Take a training sample v, compute the probabilities of the hidden units and sample a hidden activation vector h from this probability distribution. Compute the outer product of v and h and call this the positive gradient. From h, sample a reconstruction v' of the visible units, then resample the hidden activations h' from this. (Gibbs sampling step) Compute the outer product of v' and h' and call this the negative gradient. Let the update to the weight matrix W {\displaystyle W} be the positive gradient minus the negative gradient, times some learning rate: Δ W = ■ (v h T − v ′ h ′ T) {\displaystyle \Delta W=\epsilon (vh^{\mathsf {T}}-v'h'^{\mathsf {T}})} . Update the biases a and b analogously: Δ a = ■ (v − v ′) {\displaystyle \Delta a=\epsilon (v-v')} , Δ b = ■ (h − h ′) {\displaystyle \Delta b=\epsilon (h-h')} . A Practical Guide to Training RBMs written by Hinton can be found on his homepage.

Section 8

They can be trained in either supervised or unsupervised ways, depending on the task. As their name implies, RBMs are a variant of Boltzmann machines, with the restriction that their neurons must form a bipartite graph:

Section 9

P (v i k = 1 | h) = exp ■ (a i k + Σ j W i j k h j) Σ k ′ = 1 K exp ■ (a i k ′ + Σ j W i j k ′ h j) {\displaystyle P(v_{i}^{k}=1|h)={\frac {\exp(a_{i}^{k}+\Sigma _{j}W_{ij}^{k}h_{j})}{\Sigma _{k'=1}^{K}\exp(a_{i}^{k'}+\Sigma _{j}W_{ij}^{k'}h_{j})}}}

Section 10

The algorithm most often used to train RBMs, that is, to optimize the weight matrix W {\displaystyle W} , is the contrastive divergence (CD) algorithm due to Hinton, originally developed to train PoE (product of experts) models. The algorithm performs Gibbs sampling and is used inside a gradient descent procedure (similar to the way backpropagation is used inside such a procedure when training feedforward neural nets) to compute weight update. The basic, single-step contrastive divergence (CD-1) procedure for a single sample can be summarized as follows:

Section 11

P (h | v) = ∏ j = 1 n P (h j | v) {\displaystyle P(h|v)=\prod _{j=1}^{n}P(h_{j}|v)} . The individual activation probabilities are given by

Section 12

Bibliography Chen, Edwin (2011-07-18). "Introduction to Restricted Boltzmann Machines". Edwin Chen's blog. Nicholson, Chris; Gibson, Adam. "A Beginner's Tutorial for

Restricted Boltzmann Machines". Deeplearning4j Documentation. Archived from the original on 2017-02-11. Retrieved 2018-11-15.{{cite web}}: CS1 maint: bot: original URL status unknown (link) Nicholson, Chris; Gibson, Adam. "Understanding RBMs". Deeplearning4j Documentation. Archived from the original on 2016-09-20. Retrieved 2014-12-29.

Section 13

The difference between the Stacked Restricted Boltzmann Machines and RBM is that RBM has lateral connections within a layer that are prohibited to make analysis tractable. On the other hand, the Stacked Boltzmann consists of a combination of an unsupervised three-layer network with symmetric weights and a supervised fine-tuned top layer for recognizing three classes. The usage of Stacked Boltzmann is to understand Natural languages, retrieve documents, image generation, and classification. These functions are trained with unsupervised pre-training and/or supervised fine-tuning. Unlike the undirected symmetric top layer, with a two-way unsymmetric layer for connection for RBM. The restricted Boltzmann's connection is three-layers with asymmetric weights, and two networks are combined into one. Stacked Boltzmann does share similarities with RBM, the neuron for Stacked Boltzmann is a stochastic binary Hopfield neuron, which is the same as the Restricted Boltzmann Machine. The energy from both Restricted Boltzmann and RBM is given by Gibb's probability measure: $E=-\frac{1}{2}\sum_{i,j} w_{ij} s_i s_j + \sum_i \theta_i s_i$ {\displaystyle E=-{\frac {1}{2}}\sum _{i,j}{w_{ij}}{s_{i}}{s_{j}}}+\sum _{i}{\theta _{i}}{s_{i}}} . The training process of Restricted Boltzmann is similar to RBM. Restricted Boltzmann train one layer at a time and approximate equilibrium state with a 3-segment pass, not performing back propagation. Restricted Boltzmann uses both supervised and unsupervised on different RBM for pre-training for classification and recognition. The training uses contrastive divergence with Gibbs sampling: $\Delta w_{ij} = e*(p_{ij} - p'_{ij})$ The restricted Boltzmann's strength is it performs a non-linear transformation so it's easy to expand, and can give a hierarchical layer of features. The Weakness is that it has complicated calculations of integer and real-valued neurons. It does not follow the gradient of any function, so the approximation of Contrastive divergence to maximum likelihood is improvised.